STAT 429: Introduction to Time Series
Week 9 Lecture Notes

University of Illinois Urbana-Champaign

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1 Parameter Estimation in AutoRegressive Models

We shall discuss parameter estimation in $AR(p)$ models. Recall that the $AR(p)$ is given by:

$$y_t = \phi_0 + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \epsilon_t \tag{1}$$

for $t = p+1, \ldots, n$. In matrix notation,

$$Y = X\beta + \epsilon$$

where

$$Y = \begin{pmatrix} y_{p+1} \\ y_{p+2} \\ \cdot \\ \cdot \\ \cdot \\ y_n \end{pmatrix} \quad X = \begin{pmatrix} 1 & y_p & y_{p-1} & \cdot & \cdot & \cdot & y_1 \\ 1 & y_{p+1} & y_p & \cdot & \cdot & \cdot & y_2 \\ 1 & y_{p+2} & y_{p+1} & \cdot & \cdot & \cdot & y_3 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 1 & y_{n-1} & y_{n-2} & \cdot & \cdot & \cdot & y_{n-p} \end{pmatrix} \quad \beta = \begin{pmatrix} \phi_0 \\ \phi_1 \\ \phi_2 \\ \cdot \\ \cdot \\ \cdot \\ \phi_p \end{pmatrix} \quad \epsilon = \begin{pmatrix} \epsilon_{p+1} \\ \epsilon_{p+2} \\ \cdot \\ \cdot \\ \cdot \\ \epsilon_n \end{pmatrix}$$

The $AR(p)$ can be seen as a special case of the usual linear regression model where the covariate matrix X as well as the response vector y are both formed from a single data set $y_1, \ldots, y_m$.

We shall discuss estimation of the parameters $\phi_0, \ldots, \phi_p$ and σ in $AR(p)$. For simplicity, let us first assume $p = 1$ (we shall revert to the more general case later). The AR(1) model is given by:

$$y_t = \phi_0 + \phi_1 y_{t-1} + \epsilon_t \quad \text{for } t = 2, \ldots, n. \tag{2}$$

Because of the close relation between $AR(p)$ models and linear regression, let us first revisit parameter estimation in usual linear regression.

1.1 Detour: usual linear regression

The model (2) looks just like a usual regression model:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i \tag{3}$$

except we are using ϕ instead of β for the coefficients, and the index is now t as opposed to i. Let the data be denoted by $(x_i, y_i), i = 1, \ldots, m$ (m is the number of data points).

Under the assumption $\epsilon_i \overset{\text{i.i.d}}{\sim} N(0, \sigma^2)$, the likelihood for (3) is:

$$\prod_{i=1}^{m} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}\right). \tag{4}$$

Let us take a closer look as to how the likelihood (4) is actually derived. We shall examine closely the assumptions that are made in deriving (4). Our main interest is to see whether the same assumptions are still true for AutoRegression.

The data is $(x_i, y_i), i = 1, \ldots, m$ (m is the number of data points). The likelihood is the probability density function of the data treated as a function of the parameters $\theta = (\beta_0, \beta_1, \sigma)$ (σ is the standard deviation of the errors):

$$\text{Likelihood for model (3)} = f_{x_1,y_1,x_2,y_2,\ldots,x_m,y_m|\theta}(x_1, y_1, \ldots, x_m, y_m).$$

Given that the model (3) writes each y_i in terms of x_i, it makes sense to first condition on $x_1, \ldots, x_m$:

$$\text{Likelihood for model (3)} = f_{y_1,\ldots,y_m|x_1,\ldots,x_m,\theta}(y_1, \ldots, y_m) f_{x_1,\ldots,x_m|\theta}(x_1, \ldots, x_m).$$

Now we write $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ for each i to get

$$\begin{aligned}
&\text{Likelihood for model (3)} \\
&= f_{y_1,\ldots,y_m|x_1,\ldots,x_m,\theta}(y_1, \ldots, y_m) f_{x_1,\ldots,x_m|\theta}(x_1, \ldots, x_m) \\
&= f_{\beta_0+\beta_1 x_1+\epsilon_1,\ldots,\beta_0+\beta_1 x_m+\epsilon_m|x_1,\ldots,x_m,\theta}(y_1, \ldots, y_m) f_{x_1,\ldots,x_m|\theta}(x_1, \ldots, x_m) \\
&= f_{\epsilon_1,\ldots,\epsilon_m|x_1,\ldots,x_m,\theta}(y_1 - \beta_0 - \beta_1 x_1, \ldots, y_m - \beta_0 - \beta_1 x_m) f_{x_1,\ldots,x_m|\theta}(x_1, \ldots, x_m).
\end{aligned}$$

To proceed further, we assume that $\epsilon_1, \ldots, \epsilon_m$ are independent of $x_1, \ldots, x_m$ (given the parameters). This allows us to remove the conditioning on $x_1, \ldots, x_m$ in the first term above, leading to:

$$\begin{aligned}
&\text{Likelihood for model (3)} \\
&= f_{\epsilon_1,\ldots,\epsilon_m|\theta}(y_1 - \beta_0 - \beta_1 x_1, \ldots, y_m - \beta_0 - \beta_1 x_m) f_{x_1,\ldots,x_m|\theta}(x_1, \ldots, x_m).
\end{aligned}$$

We now use the assumption that $\epsilon_1, \ldots, \epsilon_m \overset{\text{i.i.d}}{\sim} N(0, \sigma^2)$ to write:

$$\begin{aligned}
&\text{Likelihood for model (3)} \\
&= \left[\prod_{i=1}^{m} f_{\epsilon_i|\theta}(y_i - \beta_0 - \beta_1 x_i)\right] f_{x_1,\ldots,x_m|\theta}(x_1, \ldots, x_m) \\
&= \left[\prod_{i=1}^{m} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}\right)\right] f_{x_1,\ldots,x_m|\theta}(x_1, \ldots, x_m).
\end{aligned}$$

How do we deal with the last term $f_{x_1,\ldots,x_m|\theta}(x_1, \ldots, x_m)$? We simply assume that this term does not depend on θ so it only becomes a constant (in terms of θ) multiplicative factor in the likelihood that can be omitted leading to:

$$\text{Likelihood for model (3)} \propto \left(\frac{1}{\sqrt{2\pi}\sigma}\right)^m \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{m}(y_i - \beta_0 - \beta_1 x_i)^2\right),$$

which coincides with (4).

To summarize, we used the following assumptions to derive the likelihood (4):

1. The model equation (3)
2. Independence of the errors $\epsilon_1, \ldots, \epsilon_m$ with the covariates $x_1, \ldots, x_m$
3. $\epsilon_i \overset{\text{i.i.d}}{\sim} N(0, \sigma^2)$

4. The density of $x_1, \ldots, x_m$ does not depend on $\theta = (\beta_0, \beta_1, \sigma)$.

Using the likelihood, frequentist inference first computes the Maximum Likelihood Estimates by maximizing the likelihood or the log-likelihood. This gives:

$$\hat{\beta} = (X^T X)^{-1} X^T y \quad \text{and} \quad \hat{\sigma}_{\text{MLE}} := \sqrt{\frac{RSS}{m}}$$

where X is the $m \times 2$ matrix with the first column consisting of all ones, and the second column consists of $x_1, \ldots, x_m$. Then the goal is to derive the distribution of the MLEs $\hat{\beta}$ and $\hat{\sigma}_{\text{MLE}}$ (given the parameters θ). For this, one again needs to use the above assumptions.

Bayesian inference proceeds by combining the prior $\beta_0, \beta_1, \log \sigma \overset{\text{i.i.d}}{\sim} \text{unif}(-C, C)$ with the likelihood to compute the posterior. In the limit $C \to \infty$, we have seen that the posterior is given by:

$$\beta \mid \text{data} \sim t_{m-2,2}\left(\hat{\beta}, \hat{\sigma}^2 (X^T X)^{-1}\right) \quad \text{where } \hat{\sigma} := \sqrt{\frac{RSS}{m-2}}.$$

A posterior for σ can also be derived (we did this in Homework one). One important point about Bayesian inference is that once the likelihood is written, we no longer care about the assumptions that were needed for writing the likelihood. Once the likelihood is written, the subsequent inference (via the posterior distribution) only uses the likelihood. In contrast, frequentist inference uses the assumptions twice: first to write the likelihood in order to compute the MLEs, and then to derive the distribution of the MLEs.

1.2 Likelihood for AR(1)

Let us now write the likelihood for the AR(1) model (2). Superficially, (2) looks the same as (3) with $i = t$, $x_i = y_{t-1}$ and $\beta_0 = \phi_0$ and $\beta_1 = \phi_1$. However, some of the other regression assumptions listed above do not hold for (2):

1. The density of $x_t = y_{t-1}$ will depend on θ (because $y_{t-1} = \phi_0 + \phi_1 y_{t-2} + \epsilon_{t-1}$ so ϕ_0, ϕ_1 and σ certainly affect y_{t-1}).

2. It is also unclear why $\epsilon_2, \ldots, \epsilon_n$ should be independent of $x_2 = y_1, \ldots, x_n = y_{n-1}$.

As a result, we cannot use the same principles as in usual linear regression to write the likelihood for AR models. Instead we shall proceed as follows (using a different set of assumptions). As the data is $y_1, \ldots, y_n$, the likelihood is given by (below $\theta = (\phi_0, \phi_1, \sigma)$ denotes the set of parameters)

$$\begin{aligned}
&\text{Likelihood for Model (2)} \\
&= f_{y_1, \ldots, y_n \mid \theta}(y_1, \ldots, y_n) \\
&= f_{y_1 \mid \theta}(y_1) f_{y_2 \mid y_1, \theta}(y_2) f_{y_3 \mid y_1, y_2, \theta}(y_3) \ldots f_{y_n \mid y_1, \ldots, y_{n-1}, \theta}(y_n) \\
&= f_{y_1 \mid \theta}(y_1) \prod_{t=2}^{n} f_{y_t \mid y_1, \ldots, y_{t-1}, \theta}(y_t) \\
&= f_{y_1 \mid \theta}(y_1) \prod_{t=2}^{n} f_{\phi_0 + \phi_1 y_{t-1} + \epsilon_t \mid y_1, \ldots, y_{t-1}, \theta}(y_t) \\
&= f_{y_1 \mid \theta}(y_1) \prod_{t=2}^{n} f_{\epsilon_t \mid y_1, \ldots, y_{t-1}, \theta}(y_t - \phi_0 - \phi_1 y_{t-1}).
\end{aligned}$$

Now we assume that ϵ_t is independent of $y_1, \ldots y_{t-1}$. This gives

$$\begin{aligned}
&\text{Likelihood for Model (2)} \\
&= f_{y_1 \mid \theta}(y_1) \prod_{t=2}^{n} f_{\epsilon_t \mid y_1, \ldots, y_{t-1}, \theta}(y_t - \phi_0 - \phi_1 y_{t-1}) = f_{y_1 \mid \theta}(y_1) \prod_{t=2}^{n} f_{\epsilon_t}(y_t - \phi_0 - \phi_1 y_{t-1}).
\end{aligned}$$

With $\epsilon_t \sim N(0, \sigma^2)$, we get

$$\text{Likelihood for Model (2)} = f_{y_1|\theta}(y_1) \prod_{t=2}^{n} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(y_t - \phi_0 - \phi_1 y_{t-1})^2\right)$$

which is equivalent to:

$$\text{Likelihood for (2)} = f_{y_1|\theta}(y_1) \left(\frac{1}{\sqrt{2\pi}\sigma}\right)^{n-1} \exp\left(-\frac{1}{2\sigma^2}\sum_{t=2}^{n}(y_t - \phi_0 - \phi_1 y_{t-1})^2\right). \quad (5)$$

To sum up, we used the following assumptions to derive the likelihood (5):

1. The model equation (2).
2. Independence of ϵ_t and $y_1, \ldots, y_{t-1}$ for each $t = 2, \ldots, n$.
3. $\epsilon_t \sim N(0, \sigma^2)$.

The likelihood (5) has the term $f_{y_1|\theta}(y_1)$ which we should make explicit before we can compute estimators. Note that the model equation (2) is only for $t = 2, \ldots, n$ which means that y_1 never appears on the left side. So it is not possible to compute $f_{y_1|\theta}(y_1)$ using (2). There are two approaches of dealing with $f_{y_1|\theta}(y_1)$.

1.2.1 Approach One: Assume $f_{y_1|\theta}(y_1)$ does not depend on θ

Here one simply assumes that $f_{y_1|\theta}(y_1)$ does not depend on θ. Then $f_{y_1|\theta}(y_1)$ becomes a constant factor in (5) that can be ignored in proportionality leading to

$$\text{Likelihood for (2)} \propto \left(\frac{1}{\sqrt{2\pi}\sigma}\right)^{n-1} \exp\left(-\frac{1}{2\sigma^2}\sum_{t=2}^{n}(y_t - \phi_0 - \phi_1 y_{t-1})^2\right). \quad (6)$$

This likelihood is identical to the likelihood in usual regression with $x_t = y_{t-1}$ (even though the assumptions that led to the likelihood are different from the case of linear regression). Because of this, Bayesian inference here (with the prior $\phi_0, \phi_1, \log\sigma \overset{\text{i.i.d}}{\sim} \text{uniform}(-C, C)$ will lead to identical results as in the case of linear regression, in the limit $C \to \infty$. In other words, the posterior for (ϕ_0, ϕ_1) will be:

$$(\phi_0, \phi_1) \mid y_1, \ldots, y_n \sim t_{n-3,2}\left((\hat{\phi}_0, \hat{\phi}_1), \hat{\sigma}^2(X^T X)^{-1}\right)$$

where $\hat{\phi}_0$ and $\hat{\phi}_1$ represent the least squares estimates obtained by regressing y_t on y_{t-1}. Note that the residual degrees of freedom (i.e., the degrees of freedom of the t-distribution) equal $n - 3$ because the number of observations in this regression equals $n - 1$ (as $t = 2, \ldots, n$) and the number of columns of X equals 2.

Frequentist inference for AutoRegression will be quite different from inference in linear regression. First note that, under the likelihood (6), the MLEs of ϕ_0, ϕ_1, σ will be identical to the MLEs in linear regression (because the likelihood (6) is the same as for linear regression). However, in order to derive the distribution of the MLEs, we now have to use the AR assumptions which are more complicated. This part is usually done by asymptotics i.e., by letting $n \to \infty$ (see for example Chapter 4 of Shumway and Stoffer, *Time Series Analysis and Its Applications*). In this analysis, inference is based on the normal distribution and justified in the large sample limit as $n \to \infty$; in other words, t-distributions no longer arise.

This is one concrete problem where Bayesian inference and frequentist inference differ. Bayesian inference is simpler while frequentist inference is more complicated and uses asymptotic arguments (specifically, Central Limit Theorems and Laws of Large Numbers for dependent random variables).

Usual library functions for AutoRegression (such as the function `AutoReg` in the `statsmodels` library) use the frequentist formulas so they give different results from those obtained by just running the OLS function for regressing y_t on $x_t = y_{t-1}$. For example, they give standard errors and z-scores as opposed to t-scores. Most of the time in practice, the difference between the two kinds of inferences is negligible. However, strictly speaking, they are different.

1.2.2 Approach Two: Computing $f_{y_1|\theta}(y_1)$ by extending (2) to $t \leq 1$

So far we assumed the model equation (2) only for $t = 2, \ldots, n$. With this, y_1 never appears on the left hand side in (2) which means that we have not assumed anything about $f_{y_1|\theta}(y_1)$. In order to be able to compute it, a natural idea is to extend the model equation for $t = 1, 0, -1, \ldots$. This allows computation of $f_{y_1|\theta}(y_1)$ in the following way. Applying (2) for $t = 1, 0, -1, -2, \ldots$ recursively, we get

$$\begin{aligned}
y_1 &= \phi_0 + \phi_1 y_0 + \epsilon_1 \\
&= \phi_0 + \phi_1 \left(\phi_0 + \phi_1 y_{-1} + \epsilon_0\right) + \epsilon_1 \\
&= \phi_0(1 + \phi_1) + \phi_1^2 y_{-1} + \phi_1 \epsilon_0 + \epsilon_1 \\
&= \phi_0(1 + \phi_1) + \phi_1^2 \left(\phi_0 + \phi_1 y_{-2} + \epsilon_{-1}\right) + \phi_1 \epsilon_0 + \epsilon_1 \\
&= \phi_0 \left(1 + \phi_1 + \phi_1^2\right) + \phi_1^3 y_{-2} + \phi_1^2 \epsilon_{-1} + \phi_1 \epsilon_0 + \epsilon_1.
\end{aligned}$$

Continuing this way with using (2) for $t = -2, -3, \ldots, -M$ (for some large M), we get

$$\begin{aligned}
y_1 &= \phi_0 \left(1 + \phi_1 + \phi_1^2 + \cdots + \phi_1^M\right) + \phi_1^{M+1} y_{-M} + \phi_1^M \epsilon_{-M+1} + \phi_1^{M-1} \epsilon_{-M+2} + \cdots + \phi_1 \epsilon_0 + \epsilon_1 \\
&= \phi_0 \sum_{j=0}^{M} \phi_1^j + \phi_1^{M+1} y_{-M} + \sum_{j=0}^{M} \phi_1^j \epsilon_{1-j}.
\end{aligned}$$

This equation is not enough to allow us to deduce $f_{y_1|\theta}(y_1)$ because it involves the unknown quantity y_{-M}. If $|\phi_1| < 1$, then the coefficient ϕ_1^{M+1} in front of y_{-M} is very small. In this case, it might make sense to ignore the term $\phi_1^{M+1} y_{-M}$ when M is large. This allows us to write

$$y_1 \approx \phi_0 \sum_{j=0}^{M} \phi_1^j + \sum_{j=0}^{M} \phi_1^j \epsilon_{1-j} \approx \phi_0 \sum_{j=0}^{\infty} \phi_1^j + \sum_{j=0}^{\infty} \phi_1^j \epsilon_{1-j} = \frac{\phi_0}{1 - \phi_1} + \sum_{j=0}^{\infty} \phi_1^j \epsilon_{1-j}.$$

The term $\sum_{j=0}^{\infty} \phi_1^j \epsilon_{1-j}$ is the sum of independent normal random variables, so it is Normal with mean zero (as each ϵ_{1-j} has mean zero) and with variance:

$$\operatorname{var}\left(\sum_{j=0}^{\infty} \phi_1^j \epsilon_{1-j}\right) = \sum_{j=0}^{\infty} \operatorname{var}\left(\phi_1^j \epsilon_{1-j}\right) = \sum_{j=0}^{\infty} \phi_1^{2j} \operatorname{var}(\epsilon_{1-j}) = \sigma^2 \sum_{j=0}^{\infty} \phi_1^{2j} = \frac{\sigma^2}{1 - \phi_1^2}.$$

Thus when $|\phi_1| < 1$, we can write

$$y_1 \sim N\left(\frac{\phi_0}{1 - \phi_1}, \frac{\sigma^2}{1 - \phi_1^2}\right).$$

which gives

$$f_{y_1|\theta}(y_1) = \frac{\sqrt{1 - \phi_1^2}}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1 - \phi_1^2}{2\sigma^2}\left(y_1 - \frac{\phi_0}{1 - \phi_1}\right)^2\right).$$

Plugging this in (5), we get

$$\begin{aligned}&\text{Likelihood for (2)}\\&= \frac{\sqrt{1-\phi_1^2}}{\sqrt{2\pi}\sigma}\exp\left(-\frac{1-\phi_1^2}{2\sigma^2}\left(y_1-\frac{\phi_0}{1-\phi_1}\right)^2\right)\\&\quad\times\left(\frac{1}{\sqrt{2\pi}\sigma}\right)^{n-1}\exp\left(-\frac{1}{2\sigma^2}\sum_{t=2}^{n}(y_t-\phi_0-\phi_1 y_{t-1})^2\right).\end{aligned}\tag{7}$$

This is a more complicated likelihood compared to (6). This is applicable only when $|\phi_1| < 1$. We shall see later the implications of this assumption.

1.3 Two AR(1) Models

We therefore have two different versions of AR(1) corresponding to each of the likelihoods (6) and (7).

The first version is:

$$\begin{aligned}&y_t = \phi_0 + \phi_1 y_{t-1} + \epsilon_t \quad \text{for } t = 2,\ldots,n\\&\epsilon_t \text{ independent of } y_1,\ldots,y_{t-1} \quad \text{for } t = 2,\ldots,n\\&\epsilon_t \overset{\text{i.i.d}}{\sim} N(0,\sigma^2) \text{ and } y_1 \text{ is a constant.}\end{aligned}$$

The second version assumes

$$y_1 = \frac{\phi_0}{1-\phi_1} + \sum_{j=0}^{\infty}\phi_1^j\epsilon_{1-j}.$$

Note that this, along with $y_2 = \phi_0 + \phi_1 y_1 + \epsilon_2$ implies that

$$y_2 = \phi_0 + \phi_1\left(\frac{\phi_0}{1-\phi_1} + \sum_{j=0}^{\infty}\phi_1^j\epsilon_{1-j}\right) + \epsilon_2 = \frac{\phi_0}{1-\phi_1} + \sum_{j=0}^{\infty}\phi_1^j\epsilon_{2-j}.$$

By induction, one can show that

$$y_t = \frac{\phi_0}{1-\phi_1} + \sum_{j=0}^{\infty}\phi_1^j\epsilon_{t-j}.$$

So the second version of the AR(1) model can be simply written as:

$$y_t = \frac{\phi_0}{1-\phi_1} + \sum_{j=0}^{\infty}\phi_1^j\epsilon_{t-j} \quad \text{for all } t = \ldots,-3,-2,-1,0,1,2,3,\ldots\tag{8}$$

where $\epsilon_t \overset{\text{i.i.d}}{\sim} N(0,\sigma^2)$. Note that this equation automatically satisfies: $y_t = \phi_0 + \phi_1 y_{t-1} + \epsilon_t$ and also that ϵ_t is independent of $y_{t-1}, y_{t-2}, \ldots$.

It is important to note that the second model (8) only makes sense when $|\phi_1| < 1$. So this second definition is not valid unless $|\phi_1| < 1$. When $|\phi_1| = 1$, one cannot make sense of the right hand side of (8) (this is because $\sum_{j=0}^{\infty}\phi_1^j\epsilon_{t-j}$ fails to converge when $|\phi_1| \geq 1$).

We shall see later that (8) is an example of a stationary model. We will explore in depth the notion of stationarity.

2 AR Models: Estimation, Inference and Prediction

The AR(p) model is given by:

$$y_t = \phi_0 + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \epsilon_t \tag{9}$$

The unknown parameters are $\phi_0, \phi_1, \ldots, \phi_p$ as well as σ (σ is the standard deviation of ϵ_t). These need to be estimated from the observed data $y_1, \ldots, y_n$.

The likelihood is (below θ denotes the vector consisting of all the parameters $\phi_0, \ldots, \phi_p$ and σ):

$$f_{y_1,\ldots,y_n|\theta}(y_1, \ldots, y_n) = f_{y_{p+1},\ldots,y_n|y_1,\ldots,y_p,\theta}(y_{p+1}, \ldots, y_n) f_{y_1,\ldots,y_p|\theta}(y_1, \ldots, y_p).$$

The first term on the right hand side above $f_{y_{p+1},\ldots,y_n|y_1,\ldots,y_p,\theta}(y_{p+1}, \ldots, y_n)$ is the conditional likelihood of $y_{p+1}, \ldots, y_n$ given $y_1, \ldots, y_p$. This conditional likelihood is calculated as

$$\begin{aligned}
&f_{y_{p+1},\ldots,y_n|y_1,\ldots,y_p,\theta}(y_{p+1}, \ldots, y_n) \\
&= \prod_{t=p+1}^{n} f_{y_t|y_{t-1},\ldots,y_1,\theta}(y_t) \\
&= \prod_{t=p+1}^{n} f_{\phi_0+\phi_1 y_{t-1}+\cdots+\phi_p y_{t-p}+\epsilon_t|y_{t-1},\ldots,y_1,\theta}(y_t) \\
&= \prod_{t=p+1}^{n} f_{\epsilon_t|y_{t-1},\ldots,y_1,\theta}(y_t - \phi_0 - \phi_1 y_{t-1} - \cdots - \phi_p y_{t-p}).
\end{aligned}$$

In order to proceed further, we shall make the following assumption:

$$\epsilon_t \mid y_{t-1}, \ldots, y_1 \sim N(0, \sigma^2) \quad \text{for each } t = p+1, \ldots, n, \tag{10}$$

which can be ensured by assuming that $\epsilon_t \sim N(0, \sigma^2)$ **and that ϵ_t is independent of** $y_1, \ldots, y_{t-1}$. With (10), we get

$$\begin{aligned}
&f_{y_{p+1},\ldots,y_n|y_1,\ldots,y_p,\theta}(y_{p+1}, \ldots, y_n) \\
&= \prod_{t=p+1}^{n} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_t - \phi_0 - \phi_1 y_{t-1} - \cdots - \phi_p y_{t-p})^2}{2\sigma^2}\right) \\
&= \left(\frac{1}{\sqrt{2\pi}\sigma}\right)^{n-p} \exp\left(-\frac{1}{2\sigma^2} \sum_{t=p+1}^{n} (y_t - \phi_0 - \phi_1 y_{t-1} - \cdots - \phi_p y_{t-p})^2\right).
\end{aligned}$$

Observe that, in order to write the above formula, we only used the model equation (9) for $t = p+1, \ldots, n$.

The conditional joint density $f_{y_{p+1},\ldots,y_n|y_1,\ldots,y_p,\theta}(y_{p+1}, \ldots, y_n)$ is called the **conditional likelihood** of the AR(p) model. The full likelihood is

$$\begin{aligned}
&f_{y_1,\ldots,y_n|\theta}(y_1, \ldots, y_n) \\
&= f_{y_{p+1},\ldots,y_n|y_1,\ldots,y_p,\theta}(y_{p+1}, \ldots, y_n) f_{y_1,\ldots,y_p|\theta}(y_1, \ldots, y_p) \\
&= \left(\frac{1}{\sqrt{2\pi}\sigma}\right)^{n-p} \exp\left(-\frac{1}{2\sigma^2} \sum_{t=p+1}^{n} (y_t - \phi_0 - \phi_1 y_{t-1} - \cdots - \phi_p y_{t-p})^2\right) f_{y_1,\ldots,y_p|\theta}(y_1, \ldots, y_p).
\end{aligned}$$

If we assume that $f_{y_1,\ldots,y_p|\theta}(y_1, \ldots, y_p)$ does not depend on θ, then maximizing the full likelihood is equivalent to maximizing the conditional likelihood.

If we want to derive $f_{y_1,\ldots,y_p|\theta}(y_1,\ldots,y_p)$ in a more principled way, then we have to use the model equation (9) for smaller values of t (i.e., $t = p, p-1, p-2, \ldots, 0, -1, \ldots$). This makes the analysis complicated and is not really worth it. It also only works under some "stationarity" assumptions on $\phi_0, \ldots, \phi_p$. It is much simpler working with the conditional likelihood.

Using the matrix notation:

$$Y_{(n-p)\times 1} = \begin{pmatrix} y_{p+1} \\ y_{p+2} \\ . \\ . \\ . \\ y_n \end{pmatrix} \qquad X_{(n-p)\times(p+1)} = \begin{pmatrix} 1 & y_p & y_{p-1} & \cdots & y_1 \\ 1 & y_{p+1} & y_p & \cdots & y_2 \\ . & . & . & \ldots & . \\ . & . & . & \ldots & . \\ . & . & . & \ldots & . \\ 1 & y_{n-1} & y_{n-2} & \cdots & y_{n-p} \end{pmatrix} \qquad \beta_{(p+1)\times 1} = \begin{pmatrix} \phi_0 \\ \phi_1 \\ . \\ . \\ . \\ \phi_p \end{pmatrix},$$

the conditional likelihood (which is also proportional to the full likelihood under the assumption that $f_{y_1,\ldots,y_p|\theta}(y_1,\ldots,y_p)$ does not depend on θ) becomes:

$$\text{likelihood} \propto \left(\frac{1}{\sqrt{2\pi}\sigma}\right)^{n-p} \exp\left(-\frac{\|Y - X\beta\|^2}{2\sigma^2}\right). \tag{11}$$

This likelihood is the same as the likelihood in linear regression with $n-p$ observations. We can therefore infer the parameters $\phi_0, \ldots, \phi_p$ and σ as in usual linear regression with the prior:

$$\phi_0, \phi_1, \ldots, \phi_p, \log\sigma \overset{\text{i.i.d}}{\sim} \text{unif}(-C, C).$$

In the limit $C \to \infty$, this allows us to write down the joint posterior of (β, σ). Integrating over σ leads to the posterior of β alone. As in ordinary linear regression, this leads to

$$\beta \mid \text{data} \sim t_{n-2p-1,p+1}\left(\hat{\beta}, \hat{\sigma}^2(X^TX)^{-1}\right)$$

where

$$\hat{\beta} := (X'X)^{-1}X'Y \quad \text{and} \quad \hat{\sigma} = \sqrt{\frac{\|Y - X\hat{\beta}\|^2}{n-2p-1}}.$$

Note that the degrees of freedom of the t-distribution above is $n-2p-1$ as the number of observations equals $n-p$ and the number of components of β is $p+1$. If inference for σ is desired, one can use:

$$\frac{\|Y - X\hat{\beta}\|^2}{\sigma^2} \mid \text{data} \sim \chi^2_{n-2p-1}.$$

Note that Bayesian inference for AR models is identical to Bayesian inference for linear regression models because the likelihood (11) is the same as in the usual linear model (with $n-p$ observations). Bayesian inference only cares about the likelihood.

Frequentist inference for the AR(p) model is based on the MLE which is given by $\hat{\beta}$ and

$$\hat{\sigma}_{\text{MLE}} = \sqrt{\frac{\|Y - X\hat{\beta}\|^2}{n-p}}.$$

To obtain frequentist confidence intervals for the parameters ϕ_i, one needs to find the distribution of $\hat{\beta}$. Here the analysis is quite different from that used in linear regression (see, for example, Section 3.5 of the book by Shumway and Stoffer titled *Time Series Analysis and its applications* (Fourth Edition)). The results turn out to be quite close to those obtained by the Bayesian method.

Unlike Bayesian inference, frequentist inference for the AR model is not identical to frequentist inference for the usual linear regression model. For example, one does not use t-distributions

for inferring the ϕ parameters in AR(p) models. Instead, one uses normal distributions (e.g., z-scores as opposed to t-scores) which are justified by asymptotic arguments that are different from and more complicated than those used for linear regression.

On the computer, estimation and inference for AR models can be done in two ways:

1. Just create y and X as above, and use OLS in statsmodels.
2. Use the AutoReg function in statsmodels.

Both methods give the same estimates of $\phi_0, \ldots, \phi_p$. The estimate of σ is slightly different: OLS gives $\hat{\sigma}_{OLS} := \sqrt{RSS/(n-2p-1)}$ and AutoReg gives $\hat{\sigma}_{MLE} := \sqrt{RSS/(n-p)}$. The two methods also give slightly different standard errors corresponding to the coefficient estimates. OLS gives square roots of the diagonal entries of $\hat{\sigma}^2_{OLS}(X^TX)^{-1}$ while AutoReg considers $\hat{\sigma}^2_{MLE}(X^TX)^{-1}$. Finally OLS does coefficient inference using the t-distribution (with $n-2p-1$ degrees of freedom) while AutoReg recommends inference using the normal distribution.

2.1 Predictions from $AR(p)$ models

If the parameters $\phi_0, \ldots, \phi_p$ and σ (collectively denoted by θ) of the AR(p) model are fixed, then the prediction for a future value y_{n+i} is given by

$$\hat{y}_{n+i}(\theta) := \mathbb{E}\left(y_{n+i} \mid y_1, \ldots, y_n, \theta\right)$$

These values are calculated recursively for $i = 1, 2, \ldots$ using the following recursion:

$$\hat{y}_{n+i}(\theta) = \phi_0 + \phi_1 \hat{y}_{n+i-1}(\theta) + \phi_2 \hat{y}_{n+i-2}(\theta) + \cdots + \phi_p \hat{y}_{n+i-p}(\theta) \quad \text{for } i = 1, 2, \ldots. \tag{12}$$

which is initialized by

$$\hat{y}_j(\theta) = y_j \quad \text{for } j = n, n-1, \ldots, n+1-p, \tag{13}$$

Since θ is unknown, we can replace it by the conditional MLE $\hat{\theta}$.

2.2 Prediction Standard Errors

To compute prediction standard errors, we can again fix the parameter values θ, and then attempt to calculate:

$$V_i(\theta) := \mathrm{var}\left(y_{n+i} \mid y_1, \ldots, y_n, \theta\right) \quad \text{for } i = 1, 2, \ldots.$$

The prediction standard error corresponding to the predicted value for y_{n+i} can then be taken to be $\sqrt{V_i(\hat{\theta})}$ (note that θ is replaced by the conditional MLE $\hat{\theta}$).

For general p, a recursion using only the variances $V_i(\theta)$ does not close without also tracking covariances. To see this, note that:

$$\begin{aligned}
V_1(\theta) &= \mathrm{var}(y_{n+1} \mid y_1, \ldots, y_n, \theta) \\
&= \mathrm{var}(\phi_0 + \phi_1 y_n + \phi_2 y_{n-1} + \cdots + \phi_p y_{n+1-p} + \epsilon_{n+1} \mid y_1, \ldots, y_n, \theta) \\
&= \mathrm{var}(\epsilon_{n+1} \mid y_1, \ldots, y_n, \theta) = \sigma^2.
\end{aligned}$$

Next

$$\begin{aligned}
V_2(\theta) &= \mathrm{var}(y_{n+2} \mid y_1, \ldots, y_n, \theta) \\
&= \mathrm{var}(\phi_0 + \phi_1 y_{n+1} + \phi_2 y_n + \cdots + \phi_p y_{n+2-p} + \epsilon_{n+2} \mid y_1, \ldots, y_n, \theta) \\
&= \mathrm{var}(\phi_1 y_{n+1} + \epsilon_{n+2} \mid y_1, \ldots, y_n, \theta) \\
&= \phi_1^2 \mathrm{var}(y_{n+1} \mid y_1, \ldots, y_n, \theta) + \mathrm{var}(\epsilon_{n+2} \mid y_1, \ldots, y_n, \theta) \\
&= \phi_1^2 V_1(\theta) + \sigma^2 = \sigma^2(1 + \phi_1^2).
\end{aligned}$$

The next term is

$$\begin{aligned} V_3(\theta) &= \text{var}(y_{n+3} \mid y_1, \ldots, y_n, \theta) \\ &= \text{var}(\phi_0 + \phi_1 y_{n+2} + \phi_2 y_{n+1} + \cdots + \phi_p y_{n+3-p} + \epsilon_{n+3} \mid y_1, \ldots, y_n, \theta) \\ &= \text{var}(\phi_1 y_{n+2} + \phi_2 y_{n+1} + \epsilon_{n+3} \mid y_1, \ldots, y_n, \theta) \\ &= \text{var}(\phi_1 y_{n+2} + \phi_2 y_{n+1} \mid y_1, \ldots, y_n, \theta) + \text{var}(\epsilon_{n+3} \mid y_1, \ldots, y_n, \theta). \end{aligned}$$

The first term in the right hand side above cannot be written down in terms of $V_1(\theta)$ and $V_2(\theta)$ alone. It also involves the covariance between y_{n+1} and y_{n+2} (given $y_1, \ldots, y_n, \theta$).

Instead of working with the variances alone, we will get the recursion by working with the conditional **covariance matrices** of $y_{n+1}, \ldots, y_{n+k}$ (given θ and the data) for $k = 1, 2, \ldots$.

Below we review some basic facts about covariance matrices.

2.2.1 Covariance Matrices

A finite number of random variables can be viewed together as a random vector. More precisely, a random vector is a vector whose entries are random variables. Let $Y = (Y_1, \ldots, Y_n)^T$ be an $n \times 1$ random vector. Its Expectation $\mathbb{E}Y$ is defined as a vector whose ith entry is the expectation of Y_i i.e., $\mathbb{E}Y = (\mathbb{E}Y_1, \mathbb{E}Y_2, \ldots, \mathbb{E}Y_n)^T$. The covariance matrix of Y, denoted by $\text{Cov}(Y)$, is an $n \times n$ matrix whose (i, j)th entry is the covariance between Y_i and Y_j. Two important but easy facts about $\text{Cov}(Y)$ are:

1. The diagonal entries of $\text{Cov}(Y)$ are the variances of $Y_1, \ldots, Y_n$. More specifically the (i, i)th entry of the matrix $\text{Cov}(Y)$ equals $var(Y_i)$.
2. $\text{Cov}(Y)$ is a symmetric matrix i.e., the (i, j)th entry of $\text{Cov}(Y)$ equals the (j, i) entry. This follows because $\text{Cov}(Y_i, Y_j) = \text{Cov}(Y_j, Y_i)$.

The following formulae are very important:

1. $\mathbb{E}(AY + c) = A\mathbb{E}(Y) + c$ for every deterministic matrix A and every deterministic vector c.
2. $\text{Cov}(AY + c) = A\text{Cov}(Y)A^T$ for every deterministic matrix A and every deterministic vector c.

As a consequence of the second formula above, we get

$$\text{var}(a^T Y) = a^T \text{Cov}(Y) a = \sum_{i,j} a_i a_j \text{Cov}(Y_i, Y_j) \quad \text{for every } n \times 1 \text{ vector } a.$$

Given two random vectors Y $(p \times 1)$ and W $(q \times 1)$, we use $\text{Cov}(Y, W)$ to denote the $p \times q$ matrix whose $(i, j)^{th}$ entry equals the covariance $\text{Cov}(Y_i, W_j)$ between Y_i and W_j. With this definition, the previous notion of $\text{Cov}(Y)$ equals simply $\text{Cov}(Y, Y)$. It can be checked that

$$\text{Cov}(AY + c, BW + d) = A\text{Cov}(Y, W)B^T.$$

2.2.2 Covariance Recursion in $AR(p)$

We shall set up a recursion for the covariance matrices:

$$\Gamma_k(\theta) := \text{Cov}\left(\begin{pmatrix} y_{n+1} \\ \cdot \\ \cdot \\ \cdot \\ y_{n+k} \end{pmatrix} \mid \theta, y_1, \ldots, y_n \right)$$

The $(i,j)^{th}$ entry of $\Gamma_k(\theta)$ is

$$\text{Cov}\left(y_{n+i}, y_{n+j} \mid y_1, \ldots, y_n, \theta\right).$$

The diagonal entries of $\Gamma_k(\theta)$ equal $V_1(\theta), \ldots, V_k(\theta)$.

To initialize the recursion for $\Gamma_k(\theta)$, note that

$$\begin{aligned}
\Gamma_1(\theta) &= \text{var}\left(y_{n+1} \mid y_1, \ldots, y_n, \theta\right) \\
&= \text{var}\left(\phi_0 + \phi_1 y_n + \ldots \phi_p y_{n+1-p} + \epsilon_{n+1} \mid y_1, \ldots, y_n, \theta\right) \\
&= \text{var}\left(\epsilon_{n+1} \mid y_1, \ldots, y_n, \theta\right) = \sigma^2
\end{aligned}$$

We now relate $\Gamma_{k+1}(\theta)$ to $\Gamma_k(\theta)$ to establish the recursion. We can write

$$\Gamma_k(\theta) = \begin{pmatrix} \Gamma_{k-1}(\theta) & \gamma_{k1}(\theta) \\ \gamma_{k1}^T(\theta) & V_k(\theta) \end{pmatrix}$$

where

$$\gamma_{k1}(\theta) := \text{Cov}\left(\begin{pmatrix} y_{n+1} \\ \cdot \\ \cdot \\ \cdot \\ y_{n+k-1} \end{pmatrix}, y_{n+k} \mid \theta, y_1, \ldots, y_n\right)$$

and, as before,

$$V_k(\theta) = \text{var}\left(y_{n+k} \mid \theta, y_1, \ldots, y_n\right).$$

We compute $\gamma_{k1}(\theta)$ as

$$\begin{aligned}
\gamma_{k1}(\theta) &:= \text{Cov}\left(\begin{pmatrix} y_{n+1} \\ \cdot \\ \cdot \\ \cdot \\ y_{n+k-1} \end{pmatrix}, y_{n+k} \mid \theta, y_1, \ldots, y_n\right) \\
&= \text{Cov}\left(\begin{pmatrix} y_{n+1} \\ \cdot \\ \cdot \\ \cdot \\ y_{n+k-1} \end{pmatrix}, \phi_0 + \phi_1 y_{n+k-1} + \phi_2 y_{n+k-2} + \cdots + \phi_p y_{n+k-p} + \epsilon_{n+k} \mid \theta, \text{data}\right) \\
&= \text{Cov}\left(\begin{pmatrix} y_{n+1} \\ \cdot \\ \cdot \\ \cdot \\ y_{n+k-1} \end{pmatrix}, \phi_1 y_{n+k-1} + \phi_2 y_{n+k-2} + \cdots + \phi_p y_{n+k-p} \mid \theta, \text{data}\right) \\
&= \text{Cov}\left(\begin{pmatrix} y_{n+1} \\ \cdot \\ \cdot \\ \cdot \\ y_{n+k-1} \end{pmatrix}, \sum_{i=1}^{k-1} a_i y_{n+i} \mid \theta, \text{data}\right)
\end{aligned}$$

where, for $i = 1, \ldots, k-1$,

$$a_i = \begin{cases} \phi_{k-i} & \text{provided } k-p \le i \le k-1 \\ 0 & \text{otherwise} \end{cases}$$

Thus if a is the $(k-1) \times 1$ vector with entries $a_1, \ldots, a_{k-1}$, we have

$$\gamma_{k1}(\theta) = \text{Cov}\left(\begin{pmatrix} y_{n+1} \\ \cdot \\ \cdot \\ \cdot \\ y_{n+k-1} \end{pmatrix}, a^T \begin{pmatrix} y_{n+1} \\ \cdot \\ \cdot \\ \cdot \\ y_{n+k-1} \end{pmatrix} \mid \theta, \text{data} \right) = \Gamma_{k-1}(\theta) a.$$

Further

$$\begin{aligned} V_k(\theta) &= \text{var}\left(y_{n+k} \mid \theta, \text{data}\right) \\ &= \text{var}\left(\phi_0 + \phi_1 y_{n+k-1} + \phi_2 y_{n+k-2} + \cdots + \phi_p y_{n+k-p} + \epsilon_{n+k} \mid \theta, \text{data}\right) \\ &= \text{var}\left(\phi_0 + \phi_1 y_{n+k-1} + \phi_2 y_{n+k-2} + \cdots + \phi_p y_{n+k-p} \mid \theta, \text{data}\right) + \sigma^2 \\ &= \text{var}\left(\sum_{i=1}^{k-1} a_i y_{n+i} \mid \theta, \text{data}\right) + \sigma^2 = a^T \Gamma_{k-1}(\theta) a + \sigma^2. \end{aligned}$$

The equation for obtaining $\Gamma_k(\theta)$ from $\Gamma_{k-1}(\theta)$ is therefore

$$\Gamma_k(\theta) = \begin{pmatrix} \Gamma_{k-1}(\theta) & \gamma_{k1}(\theta) \\ \gamma_{k1}^T(\theta) & V_k(\theta) \end{pmatrix} = \begin{pmatrix} \Gamma_{k-1}(\theta) & \Gamma_{k-1}(\theta) a \\ a^T \Gamma_{k-1}(\theta) & a^T \Gamma_{k-1}(\theta) a + \sigma^2 \end{pmatrix}$$

The algorithm for calculating the variances $V_i(\theta)$ for $i = 1, 2, \ldots, K$ is thus given by

1. Initialize with $\Gamma_1(\theta) = V_1(\theta) = \sigma^2$.
2. For $k = 2, 3, \ldots, K$, repeat the following
   a) Form the $(k-1) \times 1$ vector a whose i^{th} entry is ϕ_{k-i} if $k - p \leq i \leq k-1$ and 0 otherwise.
   b) Calculate $\Gamma_k(\theta)$ using $\Gamma_{k-1}(\theta)$ and a by the formula given above.
3. The variances $V_i(\theta), i = 1, 2, \ldots, K$ are given by the diagonal entries of the matrix $\Gamma_K(\theta)$.

Because θ is unknown, in practice, we run this recursion with θ replaced by its conditional MLE $\hat{\theta}$. The prediction standard errors are then the square roots of $V_i(\hat{\theta})$. These prediction errors coincide with those given by the `get_prediction` function in `statsmodels`.